

DPP NO.-3

- Q.1 Find the number of ways in which letters of the word VALEDICTORY be arranged so that the vowels may never be separated.
- Q.2 How many numbers between 400 and 1000 (both exclusive) can be made with the digits 2,3,4,5,6,0 if
(a) repetition of digits not allowed. (b) repetition of digits is allowed.
- Q.3 Number of odd integers between 1000 and 8000 which have none of their digits repeated, is
(A) 1014 (B) 810 (C) 690 (D) 1736
- Q.4 If ${}^{20}P_r = 13 \times {}^{20}P_{r-1}$, then the value of r is _____.
- Q.5 The number of ways in which 5 different books can be distributed among 10 people if each person can get at most one book is :
(A) 252 (B) 10^5 (C) 5^{10} (D) ${}^{10}C_5 \cdot 5!$
- Q.6 The product of all odd positive integers less than 10000, is
(A) $\frac{(10000)!}{(5000!)^2}$ (B) $\frac{(10000)!}{2^{5000}}$ (C) $\frac{(9999)!}{2^{5000}}$ (D) $\frac{(10000)!}{2^{5000} \cdot (5000)!}$
- Q.7 The 9 horizontal and 9 vertical lines on an 8×8 chessboard form 'r' rectangles and 's' squares. The ratio $\frac{s}{r}$ in its lowest terms is
(A) $\frac{1}{6}$ (B) $\frac{17}{108}$ (C) $\frac{4}{27}$ (D) none
- Q.8 There are 720 permutations of the digits 1, 2, 3, 4, 5, 6. Suppose these permutations are arranged from smallest to largest numerical values, beginning from 1 2 3 4 5 6 and ending with 6 5 4 3 2 1.
(a) What number falls on the 124th position? (b) What is the position of the number 321546?
- Q.9 A student has to answer 10 out of 13 questions in an examination . The number of ways in which he can answer if he must answer atleast 3 of the first five questions is :
(A) 276 (B) 267 (C) 80 (D) 1200
- Q.10 The number of three digit numbers having only two consecutive digits identical is
(A) 153 (B) 162 (C) 180 (D) 161
- Q.11 Number of 3 digit numbers in which the digit at hundredth's place is greater than the other two digit is
(A) 285 (B) 281 (C) 240 (D) 204
- Q.12 Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time are such that the digit
1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(e.g. 815723946 would be one such permutation)
(A) $9 \cdot 7!$ (B) $8!$ (C) $5! \cdot 4!$ (D) $8! \cdot 4!$